



6 (a) Define resistance.

.....
 [1]

(b) A cylindrical metal wire of length 2.4 m and cross-sectional area $8.0 \times 10^{-6} \text{ m}^2$ has a resistance of 0.33Ω . There is a current in the wire of 4.7 A.

(i) Determine the resistivity of the metal from which the wire is made.

resistivity = $\Omega \text{ m}$ [2]

(ii) Calculate the charge that passes through the wire in a time of 5.0 minutes.

charge = C [2]

(iii) The free electrons (charge carriers) in the wire have an average drift speed of 0.16 mm s^{-1} .

Determine the number density of charge carriers in the metal.

number density = m^{-3} [2]





- (c) The wire in (b) may be considered to be a fixed resistor. It is connected in series with a thermistor to a battery that has negligible internal resistance.
- (i) Use circuit symbols to complete Fig. 6.1 to show the circuit diagram of this arrangement.

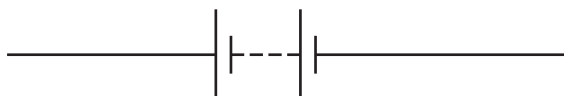


Fig. 6.1

[1]

- (ii) Explain, without calculation, how the power dissipated in the wire changes as the temperature of the thermistor is increased.

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..... [2]

[Total: 10]

